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di Torino



PIC4SeR

SENSORS, EMBEDDED SYSTEMS AND ALGORITHMS FOR SERVICE ROBOTICS (SESASR)

Robot Localization with ROS 2

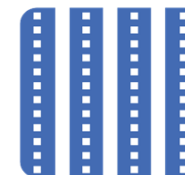
Test on real robot

Politecnico di Torino

M.Sc. Mechatronic Engineering



PRECISION
AGRICULTURE



SMART CITY
SEARCH &
RESCUE



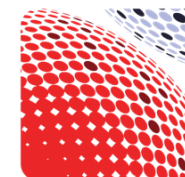
SERVICE
ROBOTICS
FOR
WELLBEING



CULTURAL
HERITAGE



UNDERWATER



SPACE



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EXPERIMENTAL SETUP

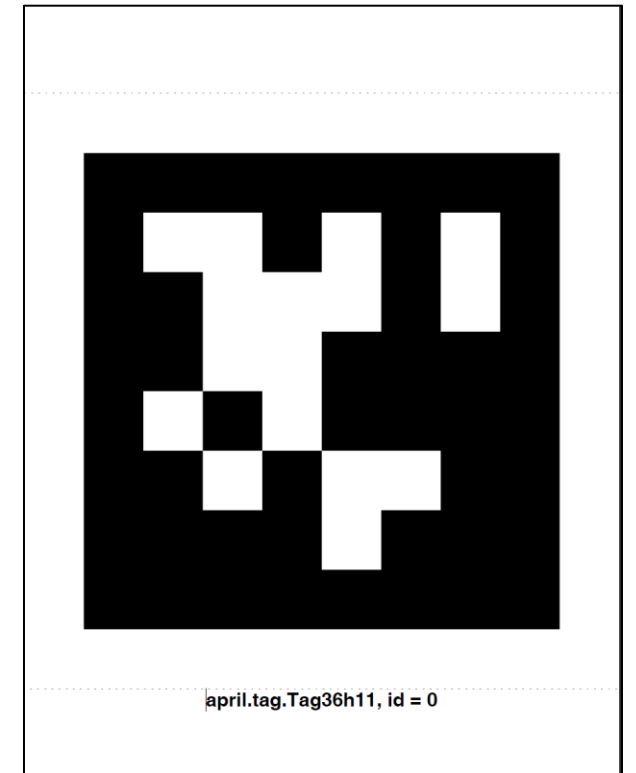
- **Robot:** Turtlebot3
- **Sensors:**
 - RGB-D Camera (RealSense D435i or Luxonis Oak-D Pro)
 - IMU
 - Odometry
- **Landmarks:**
 - AprilTags (visual fiducial markers, side length 16 cm)
 - Detected through RGB camera
 - Published as `landmark_msgs/msg/LandmarkArray`



Intel RealSense D435i



Luxonis OAK-D Pro

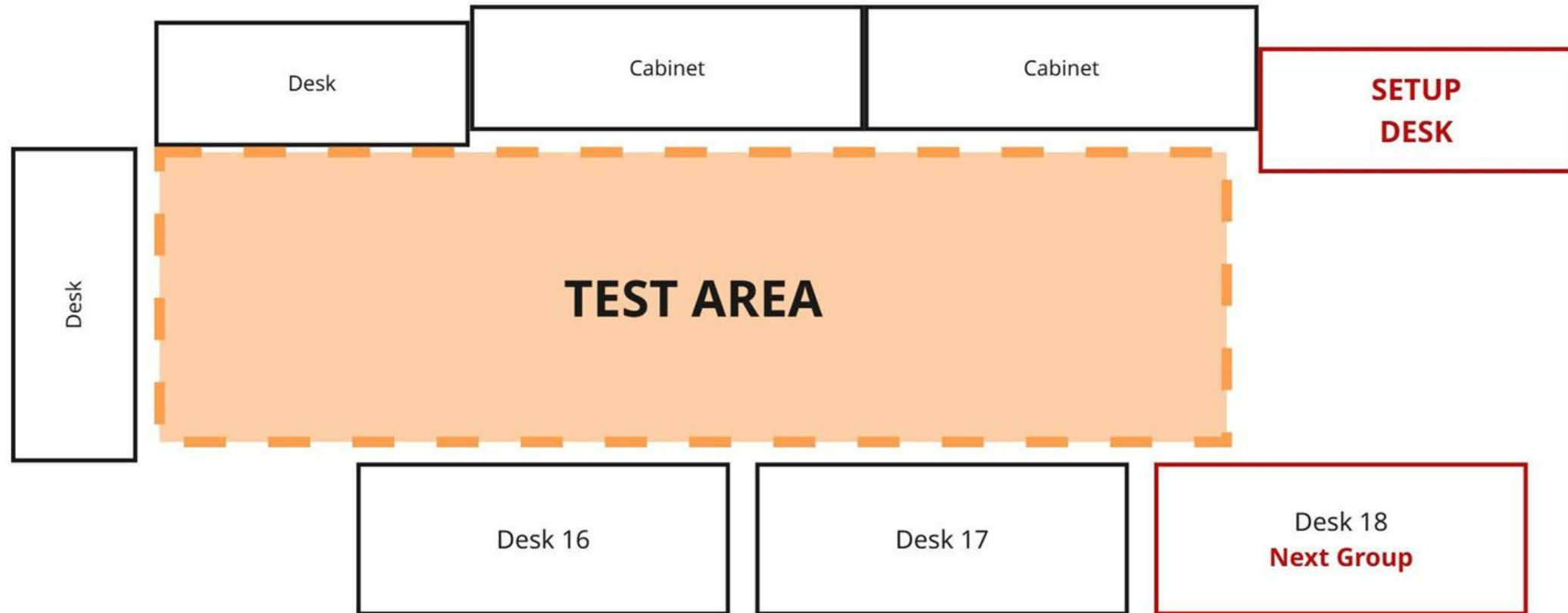




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TEST AREA ORGANIZATION





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OPERATIVE RULES

- Average test duration:

- **SETUP:** 3 min
- **EXPERIMENT:** max. 1 min



In a lab session (considering ~30 min setup)
150 min = ~**37** test slots

- Do every operation related to code editing, workspace build, parameters configuration at **your desk** using **power supply**.
- When you are at the test area you shall only **start** the necessary **nodes** and **record the data**.
- **Test area rules:**
 - The **group performing the experiment** can stay at the setup desk
 - The **next group** can stay at desk 18 preparing the robot
 - Pay attention **to not move** the **landmarks** (or some time will be lost to fix their positions)



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EXAMPLE TEST PROCEDURE

- Terminal #1 (**Robot SSH**)
 - `ros2 run rmw_zenoh_cpp rmw_zenohd`
- Terminal #2 (**Robot SSH**)
 - `ros2 launch turtlebot3_perception camera.launch.py`
- Terminal #3 (**Robot SSH**)
 - `ros2 launch turtlebot3_perception apriltag.launch.py`
- Terminal #4 (**Robot SSH or your PC**)
 - `ros2 run turtlebot3_teleop teleop_keyboard`
- Terminal #5 (**Robot SSH**)
 - `ros2 launch turtlebot3_bringup robot.launch.py`
- Terminal #6 (**Robot SSH**)
 - `ros2 bag record /cmd_vel /odom /ekf /camera/landmarks`
- Terminal #7 (**Robot SSH**)
 - `ros2 run <your_pkg_name> <your_node_name>`

Place the robot in the starting place – Press the reset button

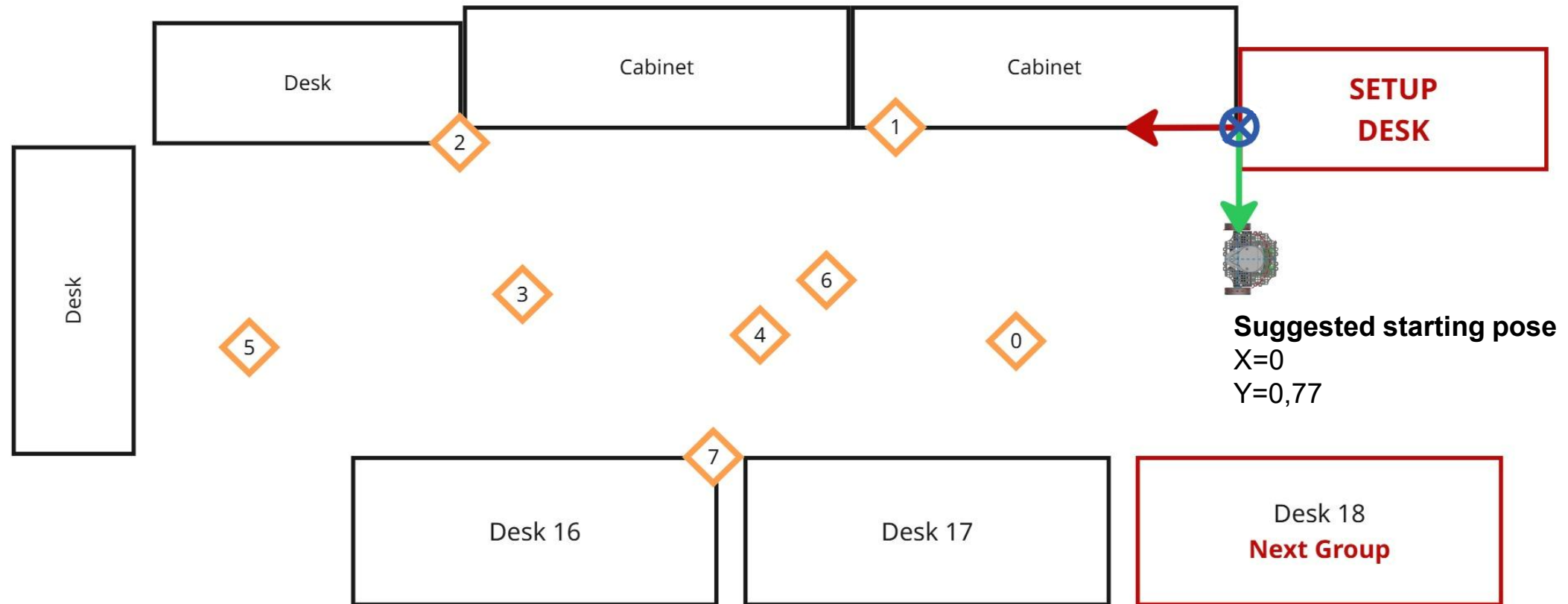
You can run the commands above the **red line** while preparing as next group



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EXPERIMENT SETUP





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LANDMARK COORDINATES

ID	x [m]	y [m]	z [m]
0	1,20	1,68	0,16
1	1,68	-0,05	0,18
2	3,72	0,14	0,22
3	3,75	1,37	0,21
4	2,48	1,25	0,22
5	4,80	1,87	0,24
6	2,18	1,00	0,24
7	2,94	2,70	0,73